

Treavor Johnson

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Summary

Engineering Specialist with **8+ years** of experience at [Denso Manufacturing](#), specializing in industrial automation, distributed systems, and manufacturing process optimization. Proficient in programming languages (Go , Python , JavaScript) and industrial controllers (Omron, Allen Bradley, Fanuc). Led capital projects worth over **\$2M**, implemented systems handling **2.7 PB** of data, and programmed **400+ machines** for data collection across seven production lines. Experience managing technical systems and managing projects with local and abroad peers. [Mechanical Engineering degree](#) with minors in Materials Science, Reliability Engineering, and Engineering Entrepreneurship.

Skills

Programming Languages: Go, Python, JavaScript, HTML, CSS, MATLAB, SQL, SASS

Web Frameworks & Tools: HTMX, Templ, Kibana, D3.js, Grafana, Vega

Databases & Storage: SQLite, Postgres, ElasticSearch

DevOps & Infrastructure: Network Management, Docker, Kubernetes, RAFT, Casbin, Gitlab CI/CD, OTEL, Ignition

Operating Systems: Linux, Windows, Container Creation

Industrial Controllers: Omron Sysmac, Omron CJ, Allen Bradley RSLogix, Allen Bradley MicroLogix, Fanuc PMC, Brother PMC

Vision Systems: Custom computer vision, Keyence, Cognex

CAD Software: SolidWorks, AutoCAD, Inventor, OnShape

Office Software: MS Outlook, MS Teams, MS Excel, MS Powerpoint, MS Project, MS Access, MS OneNote, Obsidian

Professional Experience

Denso Manufacturing Maryville, TN

Advanced Engineering Specialist (E3) | May 2017 – Present

- Program and set up SCADA systems for over **400+ machines** across seven lines
- Optimize Allen Bradley, Omron Sysmac, and Omron CJ PLC programming for cycle time and readability
- Lead multiple projects from initial design review through implementation worth over **\$2M**
- Host cross-functional teams to analyze and implement process improvements
- Analyze datasets over **100K+ data points** to ensure process reliability
- Write specifications for new machines and lines up to **\$25M**
- Create over **50 CNC programs** for aluminum machining and steel turning
- Manage network systems for industrial infrastructure and global connectivity
- Create agentic tools for automation and intelligent decision-making within Denso operations
- Manage **2.7 PB** total storage for image retention across 13 manufacturing lines

Key Technical Projects:

Casting Data Collection Platform

- Developed a production system running for over **1 year** with **zero downtime** collecting data from casting and cutting equipment
- Custom file parsers and dynamic workflow engine for external API calls
- Visual manual inspection interface with real-time data monitoring
- API layer for part traceability logging from multiple machines
- All data stored in `ElasticSearch` with automated shift report generation
- Automated log capture within `ElasticSearch` for troubleshooting and diagnostics
- Custom authentication implementation supporting username/password and access badge integration

AMR Control System

- Created a containerized control system using `Python`, `JavaScript`, `HTML`, `CSS`, `HTMX`, and PLC logic for scheduling autonomous mobile robot jobs and changing machine state
- Achieved **99.99% uptime** before project completion
- System was distributed across multiple different isolated networks with gateway devices

The University of Tennessee, Knoxville – Knoxville, TN

Undergraduate Researcher – MABE | January 2017 – May 2017

- Build implicit 2D transient multi-boundary heat transfer solver in MATLAB
- Create new tooling tripling tool life for whirling machines
- Collaborate with peers to design and validate heat transfer equipment
- Read and analyze numerous academic papers

JTEKT Automotive – Vonore, TN

Manufacturing Engineering Intern | May 2016 – January 2017

- Improve manufacturing processes to improve system reliability and cycle time
- Test machine capability to advise on new product lines and machine maintenance
- Perform ROI calculations on new automation

Waupaca Foundry Inc. – Etowah, TN

Intern/Co-op | Summers & Winters May 2013 – January 2015, May 2015 – January 2016

- Manage projects and teams of workers of over ten employees
- Design and improve parts for various types of machinery
- Produce thousands of 2-D and 3-D models for new and existing equipment
- Create and organize a file system to hold millions of files

Education

The University of Tennessee, Knoxville

Honors Bachelor of Science in Mechanical Engineering | May 2017

Minors:

- Materials Science and Engineering
- Reliability and Maintainability Engineering
- Engineering Entrepreneurship